Area, Perimeter and Volume Review

Name _____ Date _____ Calculator allowed Version 1

Scan to practise this topic online

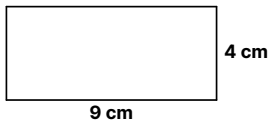
KEY WORDS area *(horahanga)* · perimeter *(paenga)* · volume *(rōrahi)* · surface area *(horahanga mata)*

A Area of flat shapes ●○○

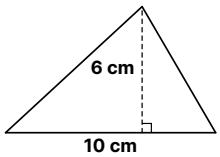
Remember: Triangle: $\frac{1}{2}bh$. Parallelogram: bh. Trapezium: $\frac{1}{2}(a+b)h$. Area is measured in squared units.

1. Find the area of each shape.

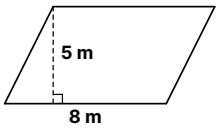
a)



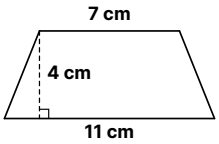
b)



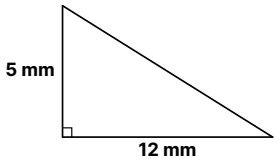
c)



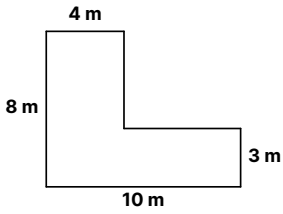
d)



e)



f)



B Perimeter and circles ●●○

Remember: Circumference $= \pi d$. Area of a circle $= \pi r^2$. Round to 1 decimal place.

2. Find the perimeter of:

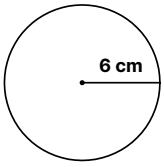
a) a rectangle 9 cm by 4 cm

b) a square with sides of 7.5 m

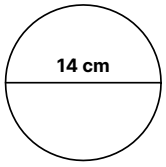
c) a regular hexagon with sides of 6 cm

3. Find the circumference of circle (a), the area of circle (b) and the area of semicircle (c).

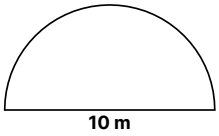
a)



b)



c)



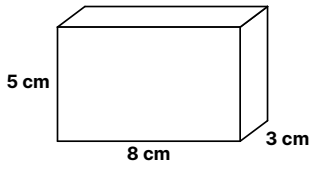
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C Volume ●●○

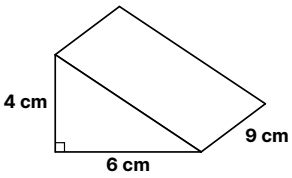
> **Remember:** Prism or cylinder: area of the end × length. Cone: $\frac{1}{3}\pi r^2 h$. Sphere: $\frac{4}{3}\pi r^3$.

4. Find the volume of each solid. Round to 1 decimal place where needed.

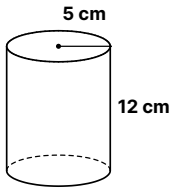
a)



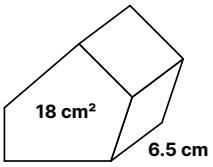
b)



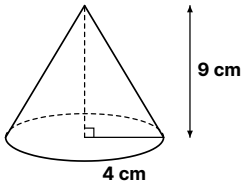
c)



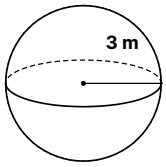
d)



e)



f)



D Surface area ●●●

> **Remember:** Find the area of every face, then add. Cylinder: $2\pi r^2 + 2\pi r h$.

5. Find the surface area of:

a) a cube with edges of 6 cm

b) a rectangular prism 10 cm by 4 cm by 3 cm

c) a cylinder with radius 3 cm and height 10 cm, to 1 decimal place

E Mixed problems ●●●

6. A rectangular paddock is 120 m long and 80 m wide. Fencing costs $14.50 for each metre. Find the cost of fencing all the way around the paddock.

7. A swimming pool is 10 m long, 4 m wide and 1.5 m deep all the way along. How many litres of water does it hold? (1 m³ = 1000 L)

8. A floor is 6 m long and 4.5 m wide. It is covered with square tiles that have sides of 0.5 m. How many tiles are needed?

9. A wheel has a radius of 25 cm. How far does it roll in 4 full turns? Give your answer in metres, to 1 decimal place.

10. A circular garden has a diameter of 5 m. Bark is spread over it 10 cm deep. Find the volume of bark needed in m³, to 2 decimal places.

★ Extension

11. A rectangle has an area of 84 cm² and a length of 12 cm. Find its perimeter.

12. A cylinder-shaped jug holds exactly 2 litres (2000 cm³). Its radius is 5 cm. Find its height, to 1 decimal place.

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Cell Tuition
1-1 PRIVATE TUTORS

GEOMETRY

Area, Perimeter and Volume Review

Name _____ **Date** _____ Calculator allowed **Version 2** Scan to practise this topic online

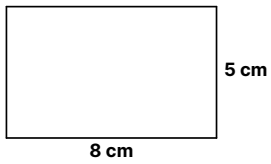
KEY WORDS area *(horahanga)* · perimeter *(paenga)* · volume *(rōrahi)* · surface area *(horahanga mata)*

A Area of flat shapes ●○○

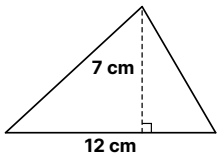
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1. Find the area of each shape.

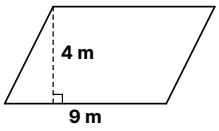
a)



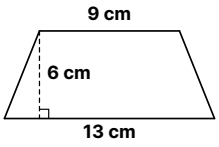
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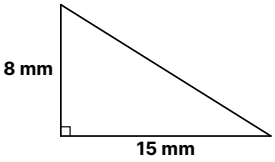
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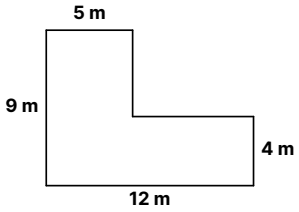
d)



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B Perimeter and circles ●●○

Remember: Circumference $= \pi d$. Area of a circle $= \pi r^2$. Round to 1 decimal place.

2. Find the perimeter of:

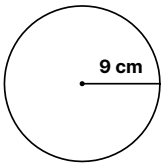
a) a rectangle 8 cm by 5 cm

b) a square with sides of 6.5 m

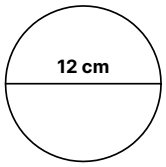
c) a regular hexagon with sides of 7 cm

3. Find the circumference of circle (a), the area of circle (b) and the area of semicircle (c).

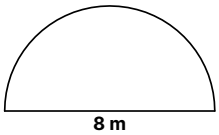
a)



b)



c)



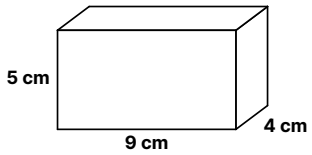
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C Volume ●●○

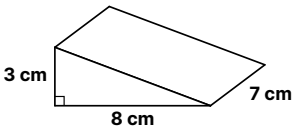
Remember: Prism or cylinder: area of the end × length. Cone: $\frac{1}{3}\pi r^2 h$. Sphere: $\frac{4}{3}\pi r^3$.

4. Find the volume of each solid. Round to 1 decimal place where needed.

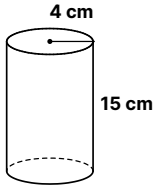
a)



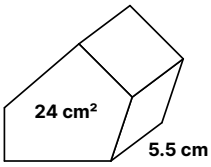
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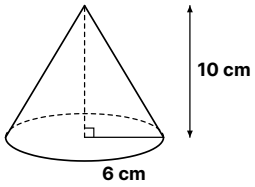
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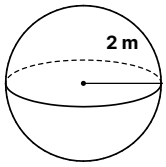
d)



e)



f)



D Surface area ●●●

Remember: Find the area of every face, then add. Cylinder: $2\pi r^2 + 2\pi r h$.

5. Find the surface area of:

a) a cube with edges of 8 cm

b) a rectangular prism 12 cm by 5 cm by 2 cm

c) a cylinder with radius 4 cm and height 9 cm, to 1 decimal place

E Mixed problems ●●●

6. A rectangular paddock is 150 m long and 90 m wide. Fencing costs $12.50 for each metre. Find the cost of fencing all the way around the paddock.

7. A swimming pool is 12 m long, 5 m wide and 1.4 m deep all the way along. How many litres of water does it hold? (1 m³ = 1000 L)

8. A floor is 8 m long and 3.5 m wide. It is covered with square tiles that have sides of 0.5 m. How many tiles are needed?

9. A wheel has a radius of 30 cm. How far does it roll in 3 full turns? Give your answer in metres, to 1 decimal place.

10. A circular garden has a diameter of 6 m. Bark is spread over it 6 cm deep. Find the volume of bark needed in m³, to 2 decimal places.

★ Extension

11. A rectangle has an area of 96 cm² and a length of 16 cm. Find its perimeter.

12. A cylinder-shaped jug holds exactly 3 litres (3000 cm³). Its radius is 6 cm. Find its height, to 1 decimal place.

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